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Cite as: Phys. Fluids **34**, 063605 (2022); doi: 10.1063/5.0093713
Submitted: 30 March 2022 · Accepted: 17 May 2022 ·
Published Online: 7 June 2022

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ABSTRACT

We numerically study the effect of the shape of frontbody and afterbody on the flow past a cylinder at a Reynolds number of 100. Two-dimensional simulations have been carried out using an in-house sharp-interface immersed boundary method-based flow solver. The cylinder cross section is considered as a semi ellipse on both windward and leeward sides. The semi-minor axis on the windward side (frontbody parameter, L_F) and the leeward side (afterbody parameter, L_A) varies from 0 to 0.5 to render cylinders of different cross sections. The effect of L_F and L_A is quantified on the following variables: drag coefficient, lift coefficient, the Strouhal number, vortex formation length, vortex fluctuation energy, the flow separation point, and cylinder bluffness. While the drag linearly decreases with both L_F and L_A, the gradient with respect to L_F is nearly twice larger than L_A. The computed vortex formation length scales directly with drag in the L_F-L_A plane, while the vortex fluctuation energy scales inversely. The lift and the Strouhal number vary non-monotonically in the L_F-L_A plane, explained in terms of vortex formation length and the flow separation point, respectively. We briefly quantify wake signatures in the L_F-L_A plane. The downstream vortex paths are traced, and in general, two vortex shedding patterns, $2S$ and $C(2S)$, are correlated with values of L_F and L_A. A dynamic mode decomposition analysis of the flow modes helps to explain the computed fluid-flow characteristics.

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I. INTRODUCTION

Vortex shedding for flow past bluff structures, such as chimneys, offshore structures, transmission cables, causes vortex-induced vibration (VIV) of the structure. Thus, the suppression of vortex shedding is a crucial factor in structural health monitoring. In this context, Rashidi *et al.*[1] summarized several active and passive vortex suppression techniques. The passive techniques include modification of roughness[2] and installing dimples,[3,4] longitudinal grooves,[5] fins,[6] or helical wire winding[7] on the cylinder surface. Vortex shedding can also be suppressed by employing attached[8,9] or detached[10–12] splitter plate, additional control cylinder,[13,14] wall proximity,[15,16] free surface proximity,[17] and slit through cavity in the cylinder.[18] Some of the active techniques for vortex suppression are via cylinder rotation,[19] stream wise oscillations,[20] transverse oscillations,[21] blowing or suction at structure surface,[22] thermal buoyancy,[23,24] and the electro-kinetic effect.[25] Chen *et al.*[26] demonstrated up to 50% lift force reduction through shape optimization of cylinder cross section.

By contrast, vortex shedding is desirable in designing VIV-based energy harvesting systems.[27] The interaction of shed vortices with a wall improves convective heat transfer[28,29] and could be useful in thermal devices. Examples include heat exchangers, flow over plates,[30] flow in pipes,[31] and cooling of I.C. engines.[32] Cylinder surface modifications like minor protrusions,[33] fins,[34] metasurfaces,[35] and thermal buoyancy[36–38] have been used to enhance vortex shedding. Islam and Mohany[39] reported the accumulation of vortex energy into lower modes with the addition of fins on a circular cylinder, increasing the vortex strength. Recently, Zhu and Liu[6] numerically studied the role of fins on force coefficients and wake structures at $Re = 60$–180. For applications in plant biomechanics, Boudina *et al.*[40] theoretically investigated capture of particles advected by flows around a cylinder at $Re \leq 300$.

Vortex shedding during flow past bluff structures has been a topic of research in the last three decades (see notable reviews 41–44). Sen *et al.*[45] studied flow past a circular cylinder at a low Reynolds number (Re) and showed that the critical Re at which shedding occurs is 46.29. A practical method to control the vortex shedding flow past a cylinder is to vary the cylinder's cross section. Several previous studies reported dependence of fluid flow characteristics and vortex shedding response on the cylinder's cross section. Bearman[41] summarized the effect of cross-section on the vortex shedding past bluff structures, namely, circular, D-section, square, and triangular cylinders, and a vertical plate. The study reported that salient sharp edges of the bluff

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structure act as flow separation points. Wahed *et al.*[46] studied flow across circular, rectangular, trapezoidal, triangular, and T-section cylinders and observed the strongest vortex shedding for the T-shaped cylinder. Sen *et al.*[47] showed that the flow past a square cylinder separates at a lower Re with a bigger wake and a larger drag, as compared to the circular cylinder. In a recent study,[48] the authors achieved a significant vortex shedding suppression by elongating the circular cylinder in the longitudinal direction of the flow. As pointed in a recent review,[49] more analysis is required to understand flow characteristics of non-circular cross sections, namely, D-section and triangular cylinders.

Contrasting from the conventional approach of studying independent shapes, some previous studies reported the effect of systematic shape variation (afterbody) using cylinders of different cross sections. Luo *et al.*[50] studied the flow across trapezoidal cylinders with a rear face size ranging from front face (square cross section) to zero (triangle cross section), at varying angles of attack. They identified two essential factors controlling vortex strength, i.e., a time delay between the vortex shedding (inverse of the Strouhal number) and the intensity of the shear layer interactions from either side. Rowe *et al.*[51] studied the effect of the boundary layer thickness on vortex shedding and observed that an increase in distance between the shear layers resulted in an increase in backpressure and a decrease in the Strouhal number. Chung and Kang[52] discussed the variation of the separation point as a controlling parameter for vortex shedding. Dobriyal *et al.*[53] studied the effects of systematically increasing afterbody from an inverted D section to elongated ellipse, keeping the frontbody identical. They observed a monotonic decrease in drag with an increase in afterbody, along with a decrease in lift. However, they reported an increase in the lift with an increase in afterbody for very small afterbody cylinders. It should be noted that these studies were primarily focused on investigating the effects of only the afterbody shape of the cylinder.

Later, some studies explored the effects of cylinder cross-section with identical frontbody and afterbody on flow characteristics. Notably, Johnson *et al.*[54,55] studied the flow past stationary bodies with systematic variation from the circular cylinder through ellipse to a vertical flat plate, using two-dimensional numerical simulations. The shape variation was quantified using the cylinder aspect ratio (the ratio of horizontal to the vertical width of the ellipse) for $Ar \in [0, 1]$. They observed that an initial Bernard von Kármán vortex shedding (2*S* pattern, i.e., two single vortices shed in an oscillation cycle) transforms to two separate coalescing vortex rows [*C*(2*S*) pattern] in the downstream flow. Further downstream flow is characterized by merging vortices and forming a secondary wake in the far downstream. Later, Radi *et al.*[56] experimentally investigated the variation of the Strouhal number with the Reynolds number (*Re*) for a similar set of elliptic cylinders. They observed the onset of three-dimensionality for cylinders with $Ar \leq 0.26$ at $Re < 100$. However, the wake re-laminarized to nearly two-dimensional flow at higher *Re*. Thompson *et al.*[57] quantified the differences in the experimental and two-dimensional simulation data of the Strouhal number and observed ∼3% variation for a vertical flat plate. They observed that the distance of secondary wake transition reduced with decreasing *Ar*; this was caused by increasing vortex flux feeding into the wake. Recently, Chen *et al.*[26] numerically demonstrated a reduction of lift coefficient by optimizing the cross-section of the cylinder.

While many studies focus on fluid flow characteristics on cylinders of different cross sections, a systematic variation of the cross-section has not been studied in detail. Specifically, those research were primarily focused on quantifying the effect of varying afterbody[50,53] or simultaneous frontbody–afterbody.[54,57] Our recent study[58] has quantified the effect of afterbody and frontbody on flow-induced vibration of an *elastically mounted* cylinder. However, to the authors' knowledge, a systematic study on an independent effect of the variation of frontbody and afterbody on a *stationary* cylinder has not been reported yet. Therefore, the present study investigates the effect of frontbody and afterbody on the variation of the following fluid flow parameters: drag and lift coefficients, the Strouhal number, vortex formation length, wake patterns, vortex strength, the separation point, and vortex paths in the downstream. Furthermore, we attempt to quantify the bluffness of various cross-sections and correlate various flow parameters with the force coefficients and the Strouhal number.

II. COMPUTATIONAL MODEL

We employed an in-house, sharp-interface immersed boundary (IB) based flow solver developed by Mittal *et al.*[59] Here, we carried out two-dimensional (2D) numerical simulations on stationary cylinders at a Reynolds number of $Re = 100$, since the flow-characteristics at $Re = 100$ are two-dimensional, in general. The 2D assumption is justified as follows. The flow past a stationary, circular cylinder remains two dimensional for $Re < 188.5$.[60] For stationary D-section cylinders, the experimental study of Boisaubert *et al.*[61] reported the existence of two-dimensional flow at $Re < 190$ for the inverted D-section and at $Re < 140$ for the D-section cylinder. Minor 3D effects may be present for the vertical plate, as reported by Thompson *et al.*[57] However, the differences in experimental and 2D simulation data of the Strouhal number are around ∼3% for the vertical plate. Sections II A–II E describe different components of the model and tests carried out in the present study.

A. Governing equations and numerical methodology

The unsteady, viscous, incompressible flow is modeled using the continuity and Navier–Stokes equations, which are expressed as follows:

$$\nabla.\mathbf{u} = 0, \tag{1}$$

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u}.\nabla)\mathbf{u} = -\nabla p + \frac{1}{Re}\nabla^2\mathbf{u}, \tag{2}$$

where non-dimensional variables are defined as follows: velocity $\mathbf{u} = \mathbf{u}^*/u_0^*$, pressure $p = p^*/\rho_f^* u_0^{*2}$, time $t = u_0^* t^*/D^*$, and the Reynolds number $Re = \rho_f^* u_0^* D^*/\mu^*$. Superscript $*$ denotes dimensional variables, and u_0^*, ρ_f^*, D^*, and μ^* are free stream velocity, fluid density, cylinder diameter, and dynamic viscosity, respectively.

Equations (1) and (2) are solved on a non-uniform Cartesian grid using a sharp-interface IB method based in-house flow solver. The IB method is well-suited to handle complex boundaries in a fluid domain and utilizes an easy-to-generate Cartesian grid for computations (see reviews 62 and 63). The present IB method is based on a multi-dimensional ghost-cell methodology developed by Mittal *et al.*[59] A detailed description of the numerical methodology is provided in previous papers.[36,59,64] Here, a brief overview of numerical methods utilized in this solver is described. The equations are discretized in

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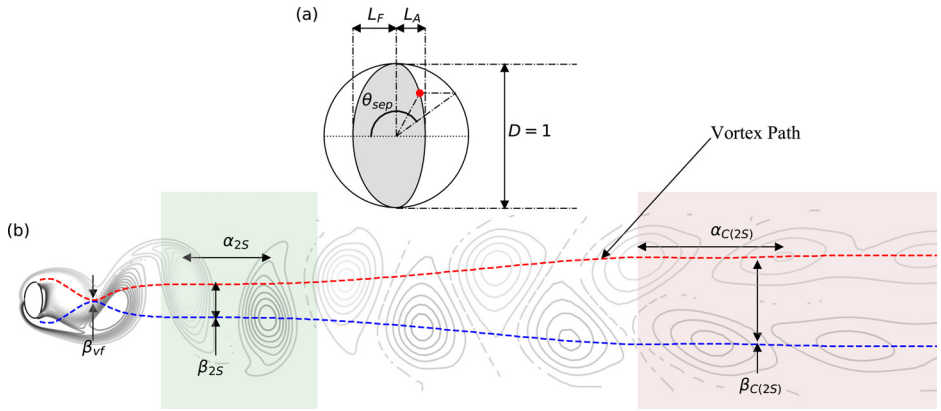


FIG. 1. (a) Definition of cross-section of the cylinder and (b) wake parameters, considered in the present study. The circumscribing circle is also shown in (a). The shaded region is the cylinder cross-section considered in the study. The subscripts *vf*, 2*S*, and *C*(2*S*) indicate the axial vortex spacing (α) and transverse vortex spacing (β) in the vortex formation, 2*S* votex shedding, and *C*(2*S*) vortex shedding regions, respectively.

space using a cell-centered, collocated (non-staggered) arrangement of velocity and pressure using the finite-difference method. The convective and diffusion terms are discretized in space using the second-order central difference method and in time using the Crank-Nicolson scheme. The time advancement is performed using the fractional step method with two steps: solving an advection-diffusion equation for an intermediate velocity field followed by a pressure Poisson equation. The pressure Poisson equation is solved using the geometric multigrid method. The obtained numerical solution is second-order accurate in space and time.[59]

B. Selection of cross sections of the cylinder and simulation setup

We consider a systematic variation of frontbody and afterbody interlinking previously studied cross sections of cylinders, namely, circular, vertical plate, D-section, and inverted D section cylinders. The cylinder cross-section is varied using frontbody (L_F)/afterbody (L_A) parameters (Fig. 1). L_F corresponds to the ratio of the semi-minor axis on the windward side with the major axis. Similarly, L_A corresponds to the ratio of the semi-minor axis on the leeward side with the major axis. The profile of the cylinder is expressed by the following equation:

$$y = \begin{cases} \pm 0.5\left[1 - \dfrac{x^2}{L_F^2}\right] & \text{for } -0.5 \le x \le 0, \\ \pm 0.5\left[1 - \dfrac{x^2}{L_A^2}\right] & \text{for } 0 \le x \le 0.5, \end{cases} \tag{3}$$

where x and y coordinates are defined with reference to the center of the cylinder. L_F and L_A are varied in the range of [0, 0.5], and the cross-sections rendered are shown on the L_F-L_A plane in Fig. 2.

A schematic of the simulation setup for the flow past the cylinder in a free stream flow is shown in Fig. 3. As discussed earlier, the simulations are performed at $Re = 100$ to avoid three-dimensional effects in the flow. As shown in Fig. 3, a uniform flow inlet boundary condition is implemented on the left wall. The right wall is imposed with a fully developed flow boundary condition. The free-slip boundary condition is imposed on the top and bottom walls. A no-slip boundary condition is imposed on the cylinder surface.

C. Parameters used to quantify the fluid flow characteristics

1. Force coefficients and the Strouhal number

The drag and lift coefficients per unit span wise length are defined as follows:

$$C_D = \frac{F_D^*}{0.5\rho_f^* u_0^{*2} D^*}, \tag{4}$$

$$C_L = \frac{F_L^*}{0.5\rho_f^* u_0^{*2} D^*}. \tag{5}$$

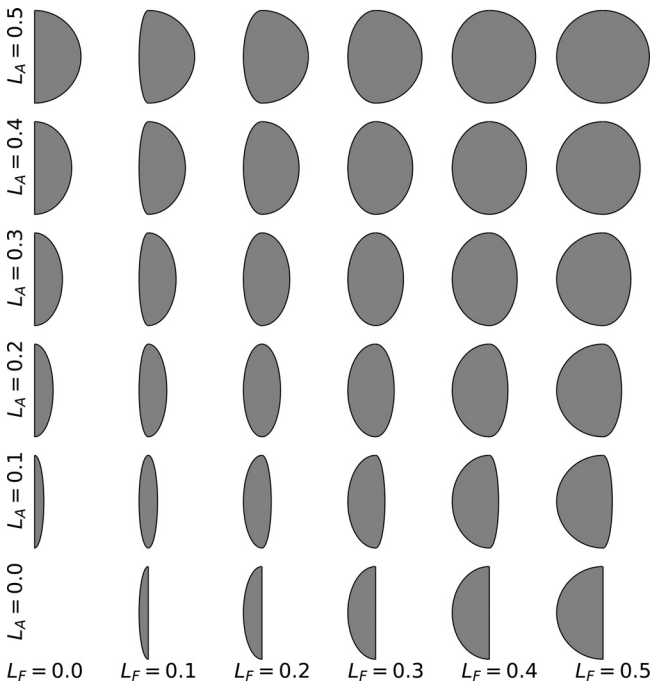


FIG. 2. Cylinders of different cross sections considered in the present study.

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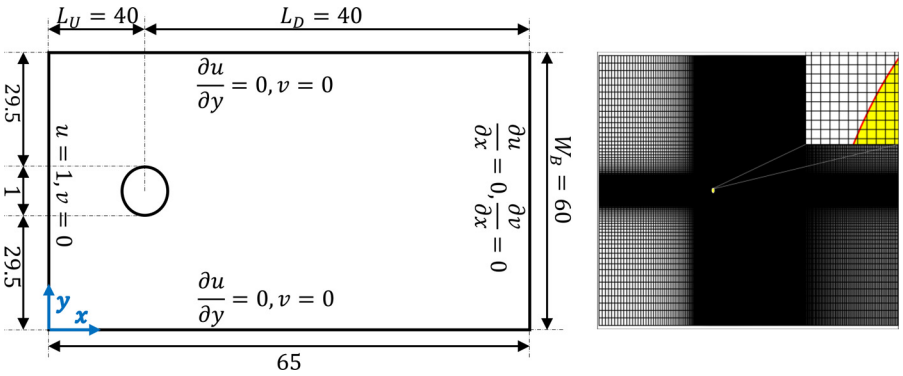


FIG. 3. Schematic of computational domain used for simulation setup and the Cartesian grid utilized in the simulations.

Mean drag C_{Dmean} and root mean square (RMS) value of lift C_{Lrms} are computed after the time-varying signals reach a dynamic steady state, and around 50 cycles are used for the averaging. The vortex shedding frequency f_v^* is non-dimensionalized as the Strouhal number

$$St = \frac{f_v^* D^*}{u_0^*}. \tag{6}$$

2. Vortex formation length

The vortex formation length $L_{\overline{e'},a}$ is used to quantify the downstream distance of vortex formation from the rearmost end of the cylinder. The definition is based on the location downstream where mean fluctuating kinetic energy is maximum anywhere in the flow. It is proposed by Chopra and Mittal[65] and is expressed as follows:

$$\overline{e'} = \frac{1}{2}; \overline{\mathbf{u}'.\mathbf{u}'} = \frac{1}{2}(\overline{u'u' + v'v'}), \tag{7}$$

where u' is defined as the transient fluctuation of the flow velocity field from the time averaged value, i.e., $u' = u - \bar{u}$. It should be noted that Chopra and Mittal[65] defined $L_{\overline{e'},a}$ as the downstream distance of the vortex formation point (location of $\overline{e'}_{max}$ in the current study) from the center of the body. However, in the present study, as the dimension of cylinders along flow direction varies significantly, we define $L_{\overline{e'},a}$ as the downstream distance of the vortex formation from the rearmost point on the cylinder.

3. Flow separation and wake quantification parameters

Figure 1(a) illustrates quantification of the separation point for different cross sections. θ_{sep} is measured as the angular location of the horizontal projection of the flow separation point (marked as a red point in Fig. 1) on the circumscribing circle to the cylinder cross section.

Figure 1(b) shows the downstream path traced by vortices, shedding from either side of the cylinder. The vortex paths correspond to the locus of the point of maximum vorticity (blue line on the lower side) and minimum vorticity (red line on the upper side), respectively. These lines are traced in the downstream only up to a location where they are well defined. The vortices are not traced in the far wake transition region, corresponding to irregular vortex shedding.[57] The inter-vortex distances α and β are defined as the in-line and transverse distance of the vortices downstream and are shown in Fig. 1(b). The subscripts *vf*, 2*S*, and *C*(2*S*) correspond to the near vortex formation point, the 2*S* vortex shedding region, and the *C*(2*S*) vortex shedding region, respectively, in the computational data (shown later). The 2*S* and *C*(2*S*) regions are highlighted using green and red backgrounds, respectively. It should be noted that the vortex shedding region is considered to be purely 2*S* or *C*(2*S*) if the vortex paths are parallel to each other.

4. Bluffness

We quantify bluffness for systematic variation in frontbody and afterbody of bluff cylinders. Roshko[66] defined bluffness as the ratio of the width of downstream wake to the width of the bluff body. Here, we quantify bluffness, $\mathbb{B}$, as the fraction of pressure component in the total drag experienced by the bluff cylinder, defined as follows:

$$\mathbb{B} = \frac{C_{Dp}}{C_D}. \tag{8}$$

The above definition is easy-to-use and consistent with the work of Griffin,[67] who defined the bluff body as a non-streamlined body. Therefore, a perfectly streamlined body will only exhibit shear drag, resulting in zero bluffness.

5. Flow modes

We employ dynamic mode decomposition (DMD) to understand the flow characteristics of different cross sections. The vorticity field is approximated by the following series summation of first L individual DMD modes:

$$\boldsymbol{\omega_z}(\boldsymbol{x}, t) = \sum_{n=-L}^{L} a_n e^{2\pi f_n t} \boldsymbol{\Phi_n}(\boldsymbol{x}), \tag{9}$$

where $\boldsymbol{\Phi_n}(\boldsymbol{x})$ is the nth spatial mode corresponding to the frequency f_n, with amplitude a_n and $a_n \geq a_{n+1}$. The modes exist in complex conjugate pairs, namely, n^{th} and $-n^{th}$ modes. We use the methodology and the algorithm described by Schmid,[68] Sayadi *et al.*,[69] and Kutz *et al.*[70] to compute the flow modes. A subdomain constituting one unit upstream, five units downstream, and two units sideways on each side from the center of the cylinder is considered to perform DMD. A uniformly spaced flow field data in time are considered for one complete

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TABLE I. Domain size independence study for grid size = 0.01. Percentage deviation is computed with respect to the biggest domain considered. The domain size parameters, W_B, and L_U and L_D, are shown in Fig. 3. CFL_{max} is the maximum CFL number in the simulation.

Domain size	W_B	L_U	L_D	CFL_{max}	C_{Dmean}	% Deviation	C_L	St
65 × 20	20	25	40	0.857	1.3735	3.22	0.231	0.1666
65 × 30	30	25	40	0.850	1.3541	1.28	0.2263	0.1648
65 × 40	40	25	40	0.848	1.3464	0.51	0.2257	0.1641
80 × 60	60	25	55	0.846	1.3413	0.0	0.2251	0.1636
65 × 60	**60**	**25**	**40**	**0.846**	**1.3412**	0.01	**0.2250**	**0.1637**
55 × 60	60	15	40	0.851	1.3559	1.46	0.2266	0.1648
50 × 60	60	10	40	0.859	1.3789	3.76	0.2329	0.1667
45 × 60	60	5	40	0.889	1.4655	12.42	0.2446	0.1739
50 × 60	60	25	25	0.846	1.3407	0.06	0.2214	0.1636
45 × 60	60	25	20	0.845	1.3401	0.12	0.2188	0.1635
40 × 60	60	25	15	0.845	1.3387	0.26	0.2151	0.1635

vortex shedding cycle, corresponding to the Strouhal frequency ($T_{DMD} = 1/St$). The transient vorticity field is segregated into distinct frequency components and associated modes. The modes are normalized such that their RSS (root sum squared) value is 1.

D. Domain and grid size independence tests

The domain-size and grid-size convergence tests are carried out for flow past a circular cylinder at $Re = 100$. The domain independence test is performed on a rectangular domain. The domain's length and width are varied from 40 to 80 in the X direction and 30–60 in the Y direction. The domain size is varied by taking three length parameters (see Fig. 3) into consideration: (i) the blockage ratio W_B, (ii) upstream non-dimensional length L_U, and (iii) downstream non-dimensional length L_D. All the lengths are non-dimensionalized using cylinder diameter D^*. One length parameter is varied, keeping all other length parameters constant. Different domains tested are listed in Table I. The minimum grid-size is kept as $\Delta x = \Delta y = 0.01$ near the cylinder for all domains. The Courant-Friedrichs-Lewy (CFL) number is kept below 1 to ensure numerical stability. Table I shows computed C_{Dmean}, C_{Lrms}, and St for different domains. The variation in results is less than 1% with respect to the results of the biggest domain considered, and therefore, a domain size of 65 × 60 is chosen for further simulations.

The grid-size independence test is performed on a non-uniform Cartesian grid with a domain of size 65 × 60. The grid distribution is kept uniform around the cylinder in a region of size $6D \times 4D$, and the minimum grid size is varied from 0.05 to 0.01 ($\Delta x = \Delta y$) to produce the test cases. We consider a sub-domain of size $2D \times 2D$ around the cylinder for computing the error norms. Figure 4 shows the variation of L_1, L_2, and L_{max} error norms for time-averaged u- and v-velocity fields on a log-log scale for different grids with respect to the finest grid. It can be observed that the error norms show nearly second-order (slope 2) convergence with the reduction in grid size. We chose the minimum grid size 0.01 (corresponding to $\Delta x = \Delta y = D/100$), with $\Delta t = 0.005$, for our further simulations since the L2 error norm is of the order of 10^{-3} in this case, and a higher accuracy can be maintained for computation of vortex formation length. Therefore, the current simulation setup, comprising a total mesh size of ∼0.5 million (1025 × 513) grid points, is well suited to simulate the flow past cylinder at $Re = 100$.

E. Verification studies

The quantification parameters used in the present study are compared with the previous studies in Table II. A non-uniform Cartesian grid with a domain size 65 × 60 and a smallest grid size 0.01 is used to simulate the flow across circular cylinder at $Re = 100$, using time step of $\Delta t = 0.005$. The computed values of C_{Dmean}, C_{Lrms}, and St show good agreement with the published ones. Additionally, the pressure component of drag C_{Dpmean}, viscous component of drag C_{Dvmean}, vortex formation length $L_{\overline{e'},a}$, vortex strength $\overline{e'}_{max}$, and the separation angle θ_{sep} are also in good agreement with the previously reported values. Finally, the DMD of flow across the circular cylinder is verified with Wang *et al.*[22] and plotted in Fig. 5. The modes are normalized for

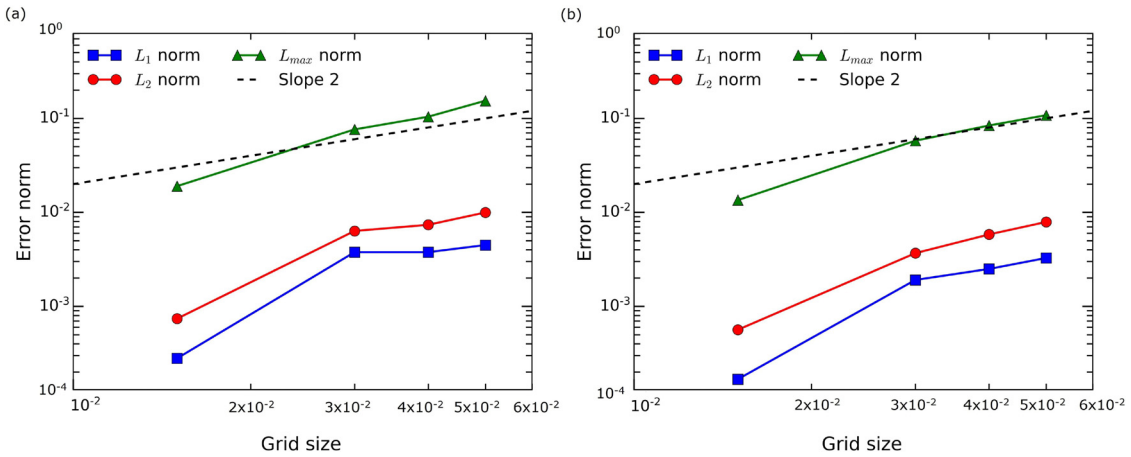


FIG. 4. Grid-size convergence study with respect to (a) u-velocity and (b) v-velocity. The dotted line represents desired second order convergence with the reduction in grid size.

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TABLE II. Computed results of verification study for domain size = 65 × 60, grid size = 0.01, and time step, $\Delta t = 0.005$. The deviation in the last row is with respect to the study of Chopra and Mittal.[65]

Study	Method	C_{Dmean}	C_{Lrms}	St	C_{Dpmean}	C_{Dvmean}	$L_{\overline{e'},a}$	$\overline{e'}_{max}$	θ_{sep}
Present	IB-FDM	**1.341**	**0.225**	**0.1637**	**1.001**	**0.3402**	**2.525**	**0.110**	**116.7°**
Chopra and Mittal[65]	GMRES-FEM	1.321	0.2238	0.1636	0.987	0.334	2.501	0.109	115.5°
Jordan and Fromm[77]	FDM	1.28	⋯	0.16	⋯	⋯	⋯	⋯	117°
Henderson[78]	Spectral element	1.351	⋯	⋯	1.003	0.351	⋯	⋯	⋯
Munday and Taira[79]	IB—projection	1.345	0.2319	0.165	⋯	⋯	⋯	⋯	⋯
Posdziech and Grundmann[80]	Spectral element	1.312	0.2235	0.1633	⋯	⋯	⋯	⋯	⋯
Deviation		1.5%	0.5%	0.06%	1.4%	1.8%	0.9%	0.9%	1.0%

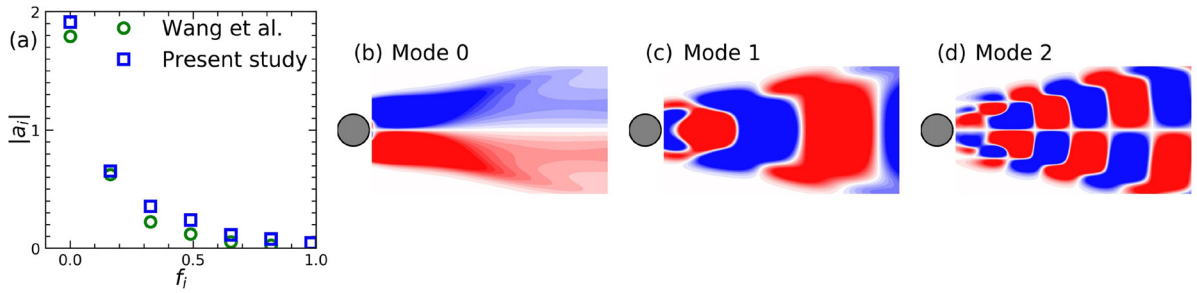


FIG. 5. (a) Comparison of DMD modal amplitude $|a_i|$ with the values reported by Wang *et al.*[71] The first three computed modes are plotted: (b) mode 0, (c) mode 1, and (d) mode 2.

domain between 0.53 and 8 units downstream of the center of the cylinder, keeping the maximum value of Φ_n as 1. The modal amplitudes and mode shapes show good agreement with Wang *et al.*[71]

III. RESULTS AND DISCUSSION

A. Force coefficients and the Strouhal number

Figures 6(a)–6(c) show the variation of mean coefficient of drag (C_{Dmean}) with frontbody (L_F), with afterbody (L_A), and on the $L_F - L_A$ plane, respectively. C_{Dmean} is the lowest for the circular cylinder and the largest on cylinders with the smallest afterbody and frontbody considered (i.e., cylinders with $L_A = 0$ and $L_F = 0.1$, and $L_F = 0$ and $L_A = 0.1$). C_{Dmean} decreases monotonically with an increase in L_F and L_A. The increase in drag with a decrease in L_A is consistent with previous reports.[50,54,57] This occurs due to a significant increase in strength of shed vortices in the downstream (quantified later in Sec. III B), attributed to an increase in curvature of the cylinder at upper and lower ends. In this context, Cheng and Liu[72] reported that the vortex strength increases with increasing sharpness of the cylinder surface while varying afterbody of a trapezoidal cylinder.

The gradient of decreasing C_{Dmean} with L_F is ∼80% higher than its gradient with L_A [Figs. 6(a) and 6(b)]. This trend is explained by reduction in static pressure with an increase in L_F due to streamlining of the cylinder windward surface. Therefore, the drag exerted on the cylinder is more sensitive to changes in frontbody than the afterbody. The variation of C_{Dmean} with L_A and L_F can be expressed using a linear least squares curve fit ($R^2 = 0.99$) as follows:

$$C_{Dmean} = 2.35 - 1.39L_F - 0.75L_A. \tag{10}$$

A dotted line in Fig. 6(c) corresponds to nearly constant C_{Dmean} for identical $1.8L_F + L_A$ cross sections and illustrates a stronger dependence of C_{Dmean} on L_F.

The variations of C_{Lrms} with L_F, with L_A, and on the $L_F - L_A$ plane are plotted in Figs. 6(d)–6(f), respectively. C_{Lrms} is the largest for the D-cylinder, while it is the lowest for a cylinder with $L_F = 0.1$ and $L_A = 0$. Cylinders with $L_A \geq 0.3$ show a monotonic decrease in C_{Lrms} with an increase in L_F [Fig. 6(d)]. This trend is consistent with the study of Luo *et al.*,[50] who showed that C_{Lrms} decreases as afterbody decreases. Authors attributed it to decreasing vortex strength and increasing vortex formation length. We quantify these variables in Sec. III B. However, C_{Lrms} increases with increase in L_A for $L_F \leq 0.2$ [Fig. 6(e)]. In this context, Dobriyal *et al.*[53] reported that with increases in L_A, C_{Lrms} increases for $0 < L_A < 0.2$ and decreases for $0.4 < L_A < 1$. Using the present data, we curve-fit C_{Lrms} as a function of L_A and L_F, and the expression obtained ($R^2 = 0.98$) is as follows:

$$C_{Lrms} = 0.16 + 0.34L_F + 1.05L_A - 0.26L_F^2 - 1.60L_FL_A - 0.79L_A^2. \tag{11}$$

The non-monotonic variation of C_{Lrms} is confirmed by Eq. (11). The coefficients of linear terms are positive, indicating a positive gradient for small values. Similarly, coefficients of quadratic terms are negative, showing a non-monotonic variation at threshold values of L_A and L_F. A dotted line in Fig. 6(f) represents local maxima of iso-value of C_{Lrms} in the L_A-L_F plane and exhibits the dependence of C_{Lrms} on both L_A and L_F.

Figures 6(g)–6(i) plot the variation of the Strouhal number (St) with L_F, with L_A, and on the $L_F - L_A$ plane, respectively. For cylinders

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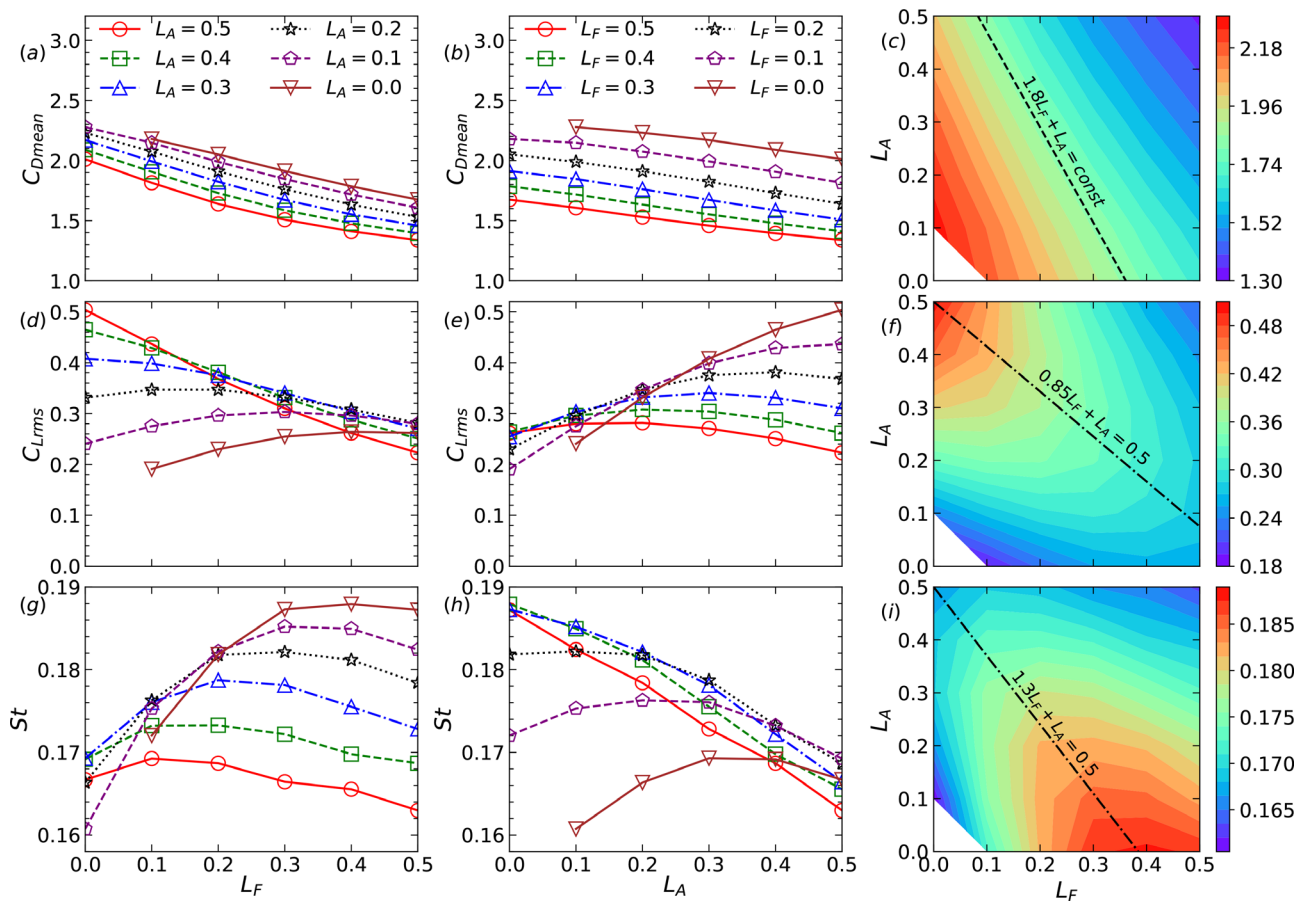


FIG. 6. Variation of C_{Dmean}, C_{Lrms}, and St with L_F (a), (d), and (g) and L_A (b), (e), and (h), respectively, while keeping the other shape parameters constant. Contour plot of iso-values of (c) C_{Dmean}, (f) C_{Lrms}, and (i) St plotted on the L_F-L_A plane. The dotted line in (c) indicates isocontours for C_{Dmean}. Similarly, the dotted lines in (f) and (i) indicate the locus of local maxima of C_{Lrms} and St on the $L_A - L_F$ plane.

with $L_A \geq 0.3$ and $L_F \geq 0.3$, St decreases with increase in L_F and L_A, respectively. Furthermore, the gradient of St with L_A is significantly larger than that with L_F. The increase in L_A and L_F results in smoother edges of the cylinder, leading to reduction in vortex strength (consistent with findings of Cheng and Liu[72]) and reduction in St (consistent with observations of Roshko[66]). However, St increases with L_F for $L_A < 0.3$ and with L_A for $L_F < 0.3$. This is explained by change in flow separation on the lee side of the cylinder (quantified in Sec. III C). A similar trend was reported by Johnson, Thompson, and Hourigan[54] and Thompson *et al.*[57] for varying elliptic cross section, keeping $L_A = L_F$. They observed an increase in St with increase in $L_A = L_F$ up to 0.25 and then decrease for $L_A = L_F > 0.25$. A detailed quantitative analysis of the primary flow parameters governing the flow dynamics is discussed in Secs. III B and III C. Using curve fitting, the variation of St with L_A and L_F can be expressed with the following fit ($R^2 = 0.94$):

$$St = 0.16 + 0.11L_F + 0.03L_A - 0.12L_F^2 - 0.13L_FL_A - 0.05L_A^2. \tag{12}$$

Similar to C_{Lrms}, St also has positive coefficients of linear terms and negative coefficients of quadratic terms, confirming the reversal in St variation after a local maxima value. We plot a dotted line in Fig. 6(f) that represents local maxima of iso-value of St in the L_A-L_F plane and exhibits the dependence of St on both L_A and L_F.

Figure 7 shows vorticity field and streamlines (first row) at instance of maximum lift coefficient. The second row plots surface pressure variation for limiting cross-section considered, namely, circular ($L_A = 0.5, L_F = 0.5$), inverted D-section ($L_A = 0.0, L_F = 0.5$), D-section ($L_A = 0.5, L_F = 0.0$), and near the flat plate ($L_A = 0.1, L_F = 0.0$). Figure 7 shows that the vortices form significantly earlier and closer to the cylinder for the near-flat plate cross-section and delayed for the circular cylinder. Therefore, the exerted pressure force on the cylinder is significantly larger for the near-flat plate cross section. Similarly, the D-section cylinder also shows a significantly closer vortex formation than the inverted D-section, resulting in higher forces exerted on the cylinder. The comparison of vortex formation length will be discussed in Sec. III B. Furthermore, the pressure force, being the dominant component in total force, can be used to qualitatively understand the C_D and C_L trends. As seen in the pressure variation of the near-flat plate, the pressure force is significantly strong. However, the direction of force is predominantly in the flow direction, and

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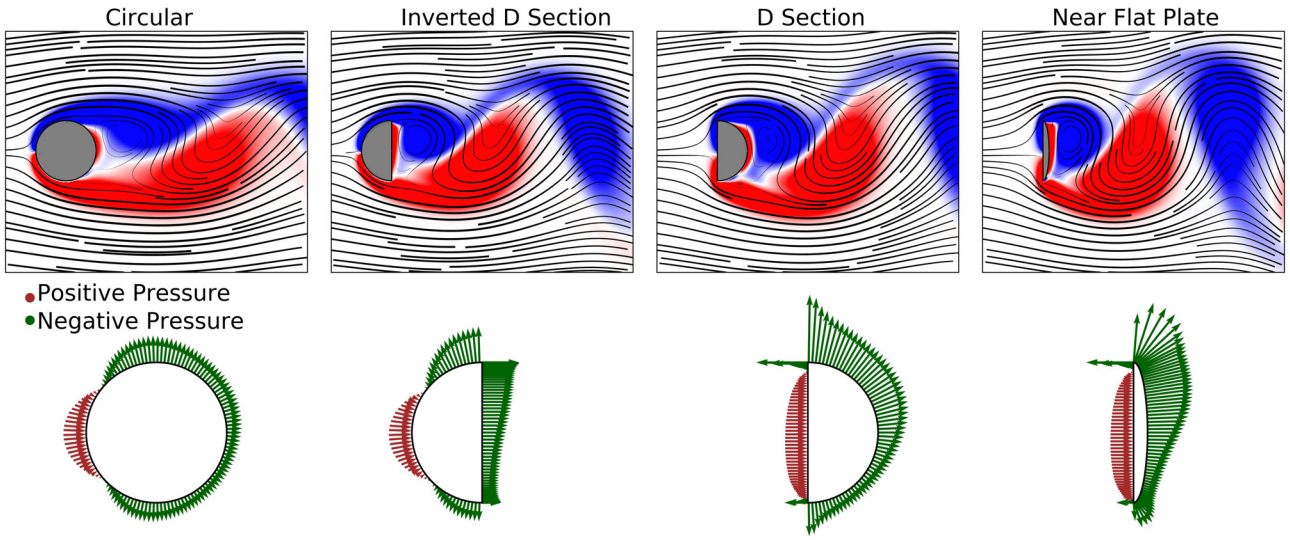


FIG. 7. Flow characteristics of limiting cases of cross-section considered: circular ($L_A = 0.5, L_F = 0.5$), inverted D-section ($L_A = 0.0, L_F = 0.5$), D-section ($L_A = 0.5, L_F = 0.0$), and the near flat plate ($L_A = 0.1, L_F = 0.0$). First row shows streamlines superimposed over the vorticity field, second row plots surface pressure distribution, and arrows show the direction of pressure force.

therefore, it only contributes toward increasing C_D and not C_L. This confirms that the increase in C_L with L_A and L_F for small $L_A - L_F$ cylinders is associated with the small transverse projected area of the cylinder. Similarly, the D-section cylinder has large pressure forces exerted by the vortices and a significant transverse projected area to transmit those forces in the transverse direction. Therefore, the D-section cylinder experiences significantly larger C_L.

B. Vortex formation length

Figures 8(a)–8(c) plot the variation of vortex formation length, $L_{\overline{e'},a}$, with L_F, with L_A, and on the $L_F - L_A$ plane, respectively. $L_{\overline{e'},a}$ monotonically increases with an increase in L_F and L_A. This trend is consistent with experimental observations by Luo *et al.*,[50] who reported a reduction in C_{Dmean} [Fig. 6(c)] and C_{Lrms} [region above dotted line in Fig. 6(f)] with an increase in the vortex formation length. Luo *et al.*[50] also reported that vortex formation length inversely scales

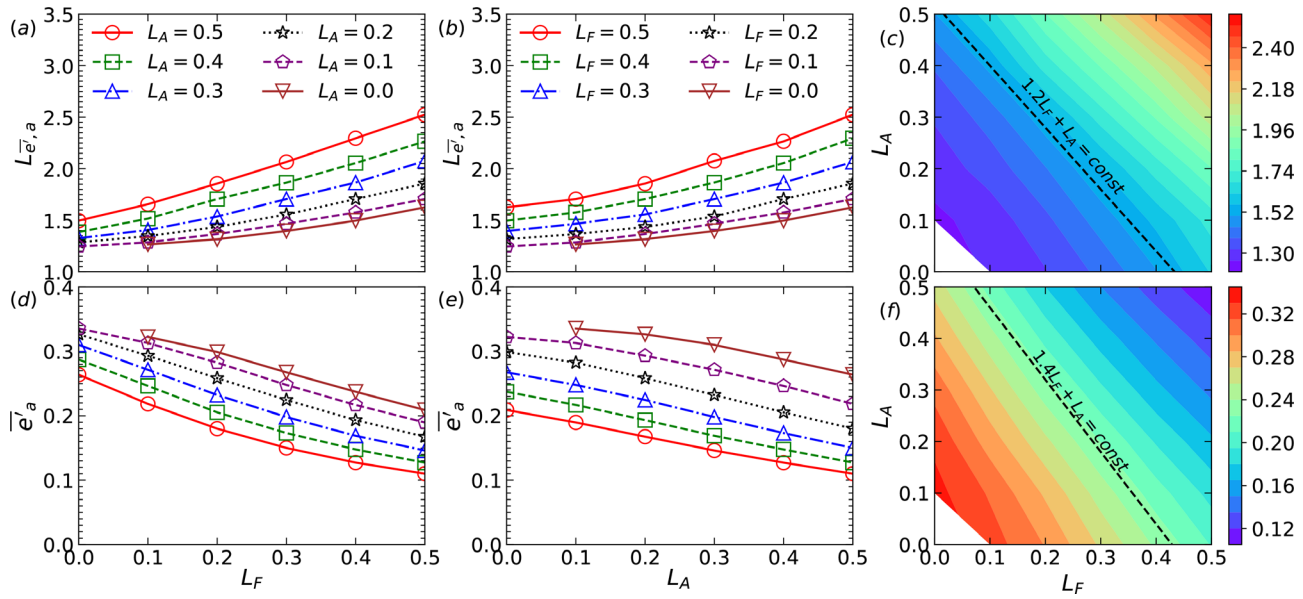


FIG. 8. Variation of vortex formation length $L_{\overline{e'},a}$ [(a) and (b)] and vortex fluctuation energy $\overline{e'}_{max}$ [(d) and (e)] with L_F and L_A of the cylinder, respectively. Contours of isovalues of (c) $L_{\overline{e'},a}$ and (f) $\overline{e'}_{max}$ in the L_F-L_F plane. Dotted lines in (c) and (f) indicate isocontours of $L_{\overline{e'},a}$ and $\overline{e'}_{max}$, respectively.

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with St for varying afterbody. In general, the present results are consistent with observations in Ref. 50 as seen in Figs. 6(b) and 8(b) for $L_F = 0$. However, for $L_F > 0.4$, St scales with $L_{\overline{e'},a}$, as seen in Figs. 6(b) and 8(b) for $L_F = 0$.

Figures 8(d) and 8(e) show the variation in the vortex strength, quantified by $\overline{e'}_{max}$, with L_F and L_A, respectively. Figure 8(f) shows variation of $\overline{e'}_{max}$ in the $L_F - L_A$ plane. $\overline{e'}_{max}$ shows a monotonic decrease with an increase in L_F and L_A. This is consistent with the observations of Thompson *et al.*,[57] who reported an increase in circulation fed to the flow for decreasing $L_F = L_A$. As seen in these figures, $\overline{e'}_{max}$ inversely scales with $L_{\overline{e'},a}$ in the $L_F - L_A$ plane. The obtained fits for $L_{\overline{e'},a}$ ($R^2 = 0.95$) and $\overline{e'}_{max}$ ($R^2 = 0.98$) variation with L_A and L_F are, respectively, expressed as follows:

$$L_{\overline{e'},a} = 1.24 + 0.50L_F + 0.42L_A, \tag{13}$$

$$\overline{e'}_{max} = 0.36 - 0.22L_F - 0.32L_A. \tag{14}$$

It should be noted that C_{Lrms} increases with increasing $L_{\overline{e'},a}$ and $\overline{e'}_{max}$ for the small L_A and L_F cylinder due to a small transverse projected area by the cylinder (see Fig. 7), and not due to the reduction in flow forces. Interestingly, the gradient of increase in $L_{\overline{e'},a}$ with L_F [see Eq. (13)] is slightly larger (~20%) as compared to that with L_A. Furthermore, the gradient of decrease in $\overline{e'}_{max}$ is ~40% higher [see Eq. (14)] for increase in L_F as compared to L_A. Therefore, the combined effect of vortex formation length and vortex strength strongly correlates with the drag force (C_{Dmean}) because it follows a similar variation with L_F and L_A.

C. Flow separation

Figures 9(a)–9(c) show the variation of flow separation angle (θ_{sep}) with L_F, with L_A, and on the $L_F - L_A$ plane, respectively. Details of computation of θ_{sep} are given in Sec. II C 2. As seen in these figures, θ_{sep} generally increases with an increase in L_F and L_A, implying a delay in flow separation. As discussed earlier, St generally decreases with an increase in L_A and L_F [see Figs. 6(c) and 6(d)]. Therefore, St decreases if θ_{sep} moves on the surface in downstream. The variation of θ_{sep} ($R^2 = 0.96$) with L_A and L_F can be expressed as follows:

$$\theta_{sep} = 90 + 13L_F + 64L_A - 17L_F^2 + 68L_FL_A - 99L_A^2. \tag{15}$$

However, for small L_F and L_A, the gradients in θ_{sep} increase. This is also evident from Eq. (15) that shows positive coefficients for linear terms and negative coefficients for quadratic terms. Specifically, as L_A approaches 0, $\theta_{sep} \to 90°$, i.e., flow separates at the top and bottom edge of the cylinder. A slight increase in L_A results in very large variation in θ_{sep} (~105°), irrespective of L_F. Interestingly, this region corresponds to the anomalous increase in St [region left to dotted line in Fig. 6(i)] with an increase in L_A. This shows that a significant shifting of the flow separation point downstream helps to increase St. These observations are consistent with $St - \theta_{sep}$ relation, reported in previous studies.[52,73]

D. Wake patterns

Figure 10 (Multimedia view) shows wake pattern for cylinders with varying L_F (along columns) and L_A (along rows). The paths followed by the vortices moving downstream are plotted using dotted lines. Details of the methodology of computing vortex paths are provided in Sec. II C. For cylinders of all cross sections considered, the circulation is fed to the vortices in a converging path along the downstream direction. The vortex paths converge near the downstream distance corresponding to the vortex formation point, which is denoted by β_{vf}, hereafter [Fig. 1(b)]. The curvature of the vortex path up to the vortex formation point inversely scales with β_{vf} and its downstream location. The curvature correlates with the shear layer interactions.[50] A smaller value of β_{vf} and the convergence of the vortex paths closer to the cylinder imply a stronger shear layer interaction. From the circular to the D-section or from the circular to the inverted D section cylinder, β_{vf} decreases and shifts upstream (Fig. 10). Therefore, as the curvature of vortex path increases, shear layer interaction becomes faster, which results in a higher St. This is consistent with results shown in Fig. 6(i) and previous findings of Luo *et al.*[50] Additionally, as stated by Miau *et al.*,[74] the curvature of vortex paths has an inverse relation with base pressure and, consequently, drag on the cylinder. A larger C_{Dmean} (see Sec. III A) of cylinders corresponds with smaller β_{vf} along with convergence of the vortex paths at smaller downstream distances [see Figs. 10 and 6(c)].

A shaded light green region in Fig. 10 corresponds to the 2S Kármán vortex street formed slightly downstream of the vortex formation point, characterized by parallel vortex paths with $\beta_{2S} \sim 1$. We define vortex street as a 2S Kármán vortex street only if the vortex paths are parallel. The β/α values in this region (see Table III) vary from 0.19 to 0.27 for cross sections considered, with higher values corresponding to small L_A and L_F. These values are very close to the theoretical predictions by Milne-Thomson,[75] who reported $\beta/\alpha \approx 0.281$

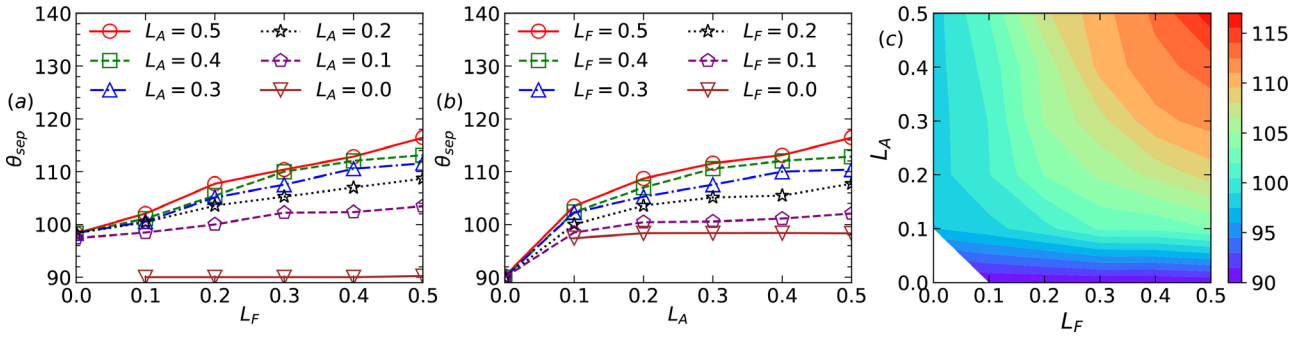


FIG. 9. Variation of separation point θ_{sep} with (a) L_F, (b) L_A, and (c) in the $L_F - L_A$ plane.

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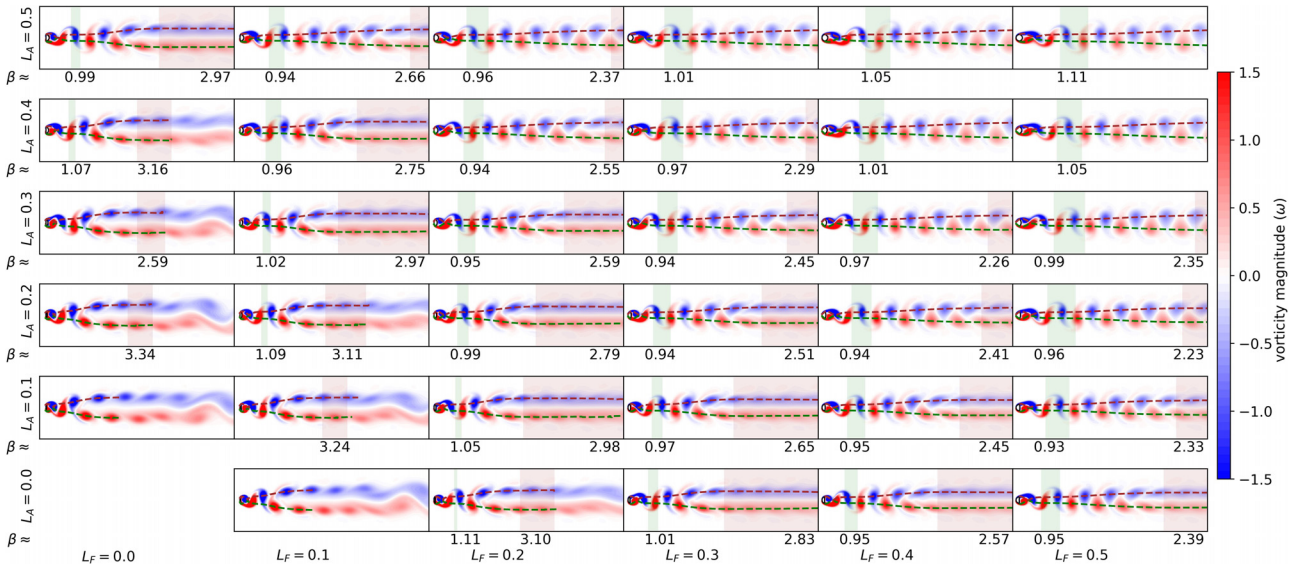


FIG. 10. Vortex shedding patterns for cross sections with varying L_F (along columns) and L_A (along rows). The common colorbar indicates the vorticity magnitude, ω for all contour plots. The vortex paths are traced using locus of vortex centers traveling downstream. The red and green shaded regions indicate 2*S* and *C*(2*S*) vortex shedding regions. The respective β values are mentioned on the lower side of each frame for 2*S* and *C*(2*S*) regions. Multimedia view: https://doi.org/10.1063/5.0093713.1.

for a stable vortex street for potential flow. This is expected, as the larger $L_A - L_F$ shapes have higher viscous components in the flow force. Similarly, the small $L_A - L_F$ shapes have a higher pressure component in the flow forces. The small $L_A - L_F$ cylinders also have a larger β value, corresponding to Roshko[66] definition of bluffness. Therefore, we have quantified bluffness based on the pressure component in flow forces in Sec. III E. It should be noted that this region is not stable for viscous flow,[76] and wake transition occurs into outward inclined vortex paths. In fact, for shapes with very small L_F and L_A, the 2*S* vortex street is never attained, and the wake directly transitions to the outward inclined vortex paths.

The further downstream flow is characterized by the formation of a *C*(2*S*) vortex shedding pattern that is highlighted in a light red background in Fig. 10 and is accompanied by parallel vortex paths at distance $\beta_{C(2S)} \in [2.29, 3.54]$. The vortex street is considered *C*(2*S*) vortex shedding if the vortex paths are parallel to each other. The *C*(2*S*) region also exists for large L_A or L_F cylinders; however, the location is much farther downstream (>100 units for a circular cylinder[57]), and the present study analyzes only up to 30 units downstream length of the cylinder. This modified vortex shedding pattern is associated with β/α (see Table IV) ranging from ∼0.478 ($L_F = 0.5, L_A = 0.3$) to ∼0.84 ($L_F = 0.1, L_A = 0.1$). The present computed results show that β/α is more sensitive to L_F.

TABLE III. Computed values of β/α corresponding to 2*S* vortex shedding pattern.

L_A \ L_F	0.0	0.1	0.2	0.3	0.4	0.5
0.5	0.205	0.189	0.188	0.193	0.197	0.203
0.4	0.235	0.204	0.194	0.194	0.197	0.201
0.3	···	0.230	0.206	0.198	0.198	0.196
0.2	···	0.254	0.227	0.208	0.201	0.199
0.1	···	···	0.251	0.225	0.214	0.201
0.0	···	···	0.274	0.240	0.217	0.212

TABLE IV. Computed values of β/α corresponding to *C*(2*S*) vortex shedding pattern.

L_A \ L_F	0.0	0.1	0.2	0.3	0.4	0.5
0.5	0.721	0.607	0.515	···	···	···
0.4	0.784	0.670	0.587	0.500	···	···
0.3	0.806	0.751	0.642	0.578	0.501	0.478
0.2	0.821	0.798	0.700	0.628	0.580	0.495
0.1	···	0.840	0.781	0.686	0.619	0.558
0.0	···	···	0.797	0.734	0.665	0.604

E. Bluffness

Figures 11(a) and 11(b) show the bluffness $\mathbb{B}$ (defined in Sec. II C 4) variation with L_F and L_A, respectively. Figure 11(c) shows the variation of $\mathbb{B}$ on the $L_F - L_A$ plane. The increase in L_F results in a significant reduction in $\mathbb{B}$. Similarly, a minor decrease in $\mathbb{B}$ is observed for increasing L_A. As observed in Fig. 11, bluffness $\mathbb{B}$ has a dominant gradient with L_F and a negligible gradient with L_A. This is possibly caused due to separated flow on the afterbody side. As the flow separates from the cylinder, the surface pressure on the lee side of the cylinder becomes nearly constant.[67] For instance, the coefficient of pressure plots by Chopra and Mittal[65] for a circular cylinder shows minor variations on the surface on the lee side past flow separation. The downstream movement of θ_{sep} with increasing L_A contributes to the minor gradient in bluffness with L_A. The vertical plate has no shear component in the drag coefficient as per the present definition of

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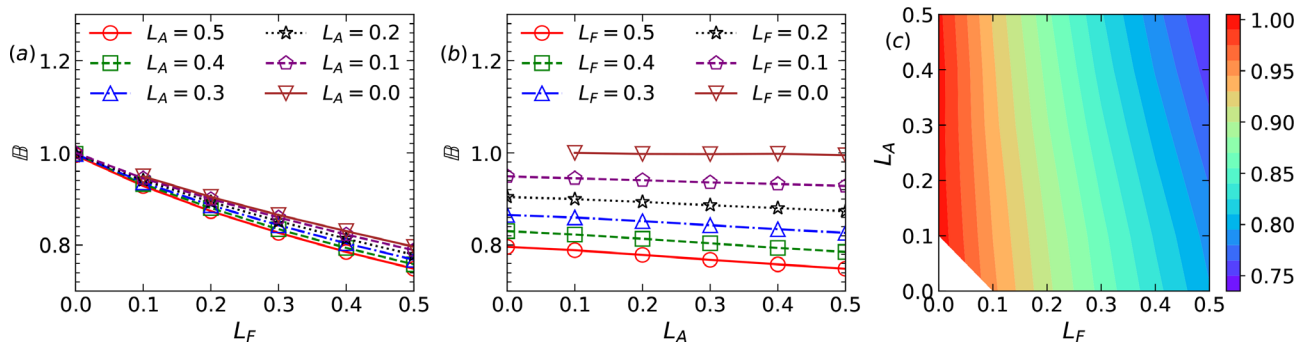


FIG. 11. Variation of bluffness of the cylinder, $\mathbb{B}$ with (a) L_F, (b) L_A, and (c) in the $L_F - L_A$ plane.

bluffness. Therefore, its theoretical bluffness value is 1. The same value is reflected in Fig. 11.

This is in accordance with the widely used bluffness definition proposed by Roshko.[66] Roshko[66] defined bluffness as the wake width (β in the present study) with respect to the frontal width of the cylinder (1 unit for all cylinders in the current study). Figure 10 shows an increase in $\beta_{C(2S)}$ with decreasing L_F, indicating an increase in bluffness for decreasing L_F. Moreover, the increase in $\beta_{C(2S)}$ (at a relatively low gradient) with a decrease in L_A is also consistent with the present bluffness definition. The computed values of β are presented only up to the second decimal place due to limitations on computational accuracy posed by the used grid size. However, Roshko's method requires accurate determination of the free field velocities. Moreover, Fig. 10 shows that cylinders with very small L_A and L_F may possibly never form a pure 2S or C(2S) vortex street and directly undergo complex far wake formation.[57] In these scenarios, the present method is relatively easy to apply, as it only requires localized force computations on the cylinder surface.

F. Characterization of flow modes by dynamic mode decomposition (DMD)

Figure 12 shows DMD of the vorticity field for flow across various cross sections. The details of the methodology are discussed in Sec. II C 5. The modal amplitude $|a_n|$ corresponds to the nth modal amplitude. Each of the modes occur at harmonics of St, i.e., $f_n = nSt$, where n is an integer. Therefore, the zeroth mode corresponds to the time-averaged flow, with further modes 1, 2, 3, 4, and 5 corresponding to the transient vortex shedding. Each sub-figure has three insets, indicating mode 0 (upper inset), mode 1 (middle inset), and mode 3 (lower inset) for the corresponding cross sections. All modes appear in conjugate pairs (Φ_n and Φ_{-n}) with equal amplitude, which is a result of symmetric vortex shedding. We analyze only near wake characteristics, and therefore, the intermediate modes corresponding to far wake characteristics, for example, as observed by Thompson *et al.*,[57] are not considered in the present analysis.

The modal amplitudes $|a_0|$, plotted in Fig. 12, show that the strength of mode 0 remains nearly constant for all cross sections. The corresponding mode shapes (upper inset) indicate that while the strengths of modes are similar, the surface curvature leads to a higher component of time averaged force in the flow direction for smaller L_A and L_F. Therefore, C_{Dmean} is higher for smaller L_A and L_F (see Fig. 6). Moreover, the mode 0 indicates a more concentrated vorticity field with decreasing L_F, which is in agreement with higher sensitivity of C_{Dmean} with L_F. $|a_1|$ corresponds to mode 1 (lower inset) oscillating at $f_1 = St$. Therefore, it is a primary indicator of C_{Lrms} and other vortex strength parameters[22] ($L_{\overline{e'},a}$ and $\overline{e'}_{max}$). Figure 12 shows a significant increase in $|a_1|$ (strength of mode 1) with decreasing L_A and L_F. Furthermore, the mode shapes of mode 1 show a decreasing contribution of vortex forces in mode 1, in the transverse direction.

Figure 13 shows the variation of mode amplitudes $|a_0|$, $|a_1|$, and $|a_2|$ with $L_A + L_F$. The error bars indicate the variation of DMD modes across shapes of identical $L_A + L_F$. A small error bar variation of $|a_0|$, $|a_1|$, and $|a_2|$ indicates a nearly constant DMD amplitude for shapes with equal $L_A + L_F$. The significantly small gradients in $|a_0|$ and $|a_2|$ with L_A and L_F indicate the smaller dependence of strength of modes 0 and 2 on the cross sections. However, $|a_1|$ shows a significant reduction with increasing $L_A + L_F$. As this mode corresponds to $f_1 = St$, it is directly correlated with the vortex formation strength. Therefore, the reduction in $|a_1|$ is consistent with the observed reduction in vortex strength, $\overline{e'}_{max}$, observed in Sec. III B.

IV. CONCLUSIONS

We have investigated the effect of frontbody and afterbody on the vortex shedding and flow force characteristics on the cylinder placed in a cross-flow at $Re = 100$. An in-house sharp-interface immersed boundary method solver is utilized to solve the two-dimensional, incompressible and viscous Navier–Stokes equations. We consider the cylinder's cross-section as a semi ellipse on both windward and leeward sides. We vary the semi-minor axis on the windward side (frontbody parameter, L_F) and leeward side (afterbody parameter, L_A) from 0 to 0.5 to render cylinders of different cross sections. Re is based on the major axis of the ellipse and is kept constant at 100 for all cross sections.

The present study reports the effect of L_F and L_A on force coefficients, C_{Dmean} and C_{Lrms}. C_{Dmean} monotonically decreases with increasing L_A and L_F, with a larger (~80%) dependence on L_F. The vortex strength, $\overline{e'}_{max}$, and vortex formation length, $L_{\overline{e'},a}$, also indicate monotonic increase and decrease, respectively, with increasing L_A and L_F. Furthermore, the gradients are ~20% and ~40% higher with L_F for $L_{\overline{e'},a}$ and $\overline{e'}_{max}$, respectively. C_{Lrms} predominantly shows a decrease with increase in L_A and L_F for large frontbody and afterbody regions. However, the trend is reversed for cross sections with L_A and L_F smaller than a threshold value. This is due to smaller transverse

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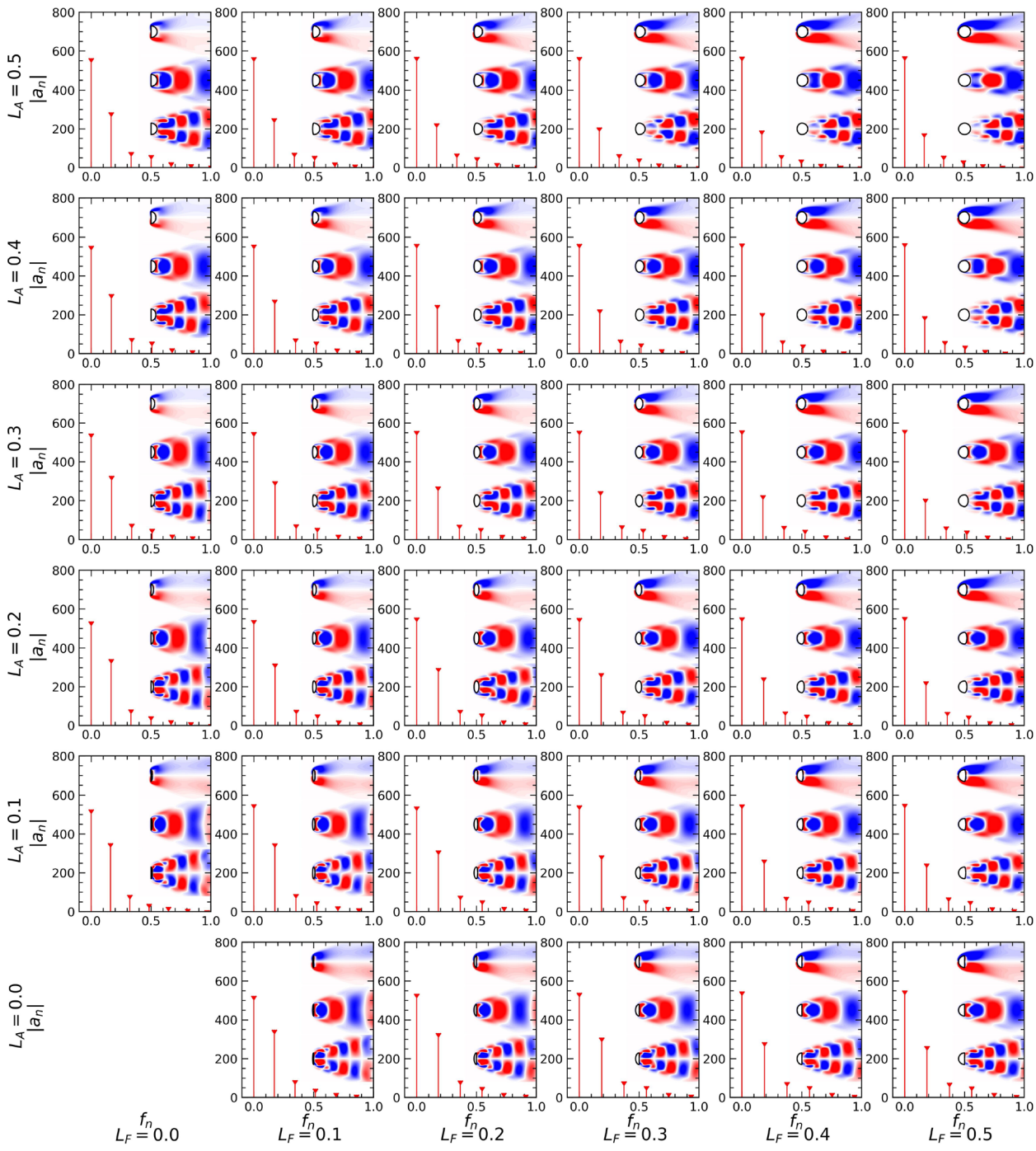


FIG. 12. Flow mode amplitudes $|a_n|$ with f_n for each cross-section with varying L_A and L_F computed using dynamic mode decomposition. Insets in each panel show mode shape of mode 0 (top inset), mode 1 (middle inset), and mode 2 (bottom inset).

projected cross section, as compared to the increase in $\overline{e}_{max}$, with decrease in L_A and L_F. This hypothesis is further strengthened by the surface pressure distribution on extreme cross-section shapes and shape of mode 1 obtained in the Dynamic mode decomposition (DMD) analysis of cross sections. *St* decreases for small $L_A + L_F$ and increases for large $L_A + L_F$, with an increase in L_A and L_F. This is caused due to a competition between $L_{\overline{e},a}$ and θ_{sep} controlling the vortex shedding frequency. The region of large $L_A + L_F$ has larger

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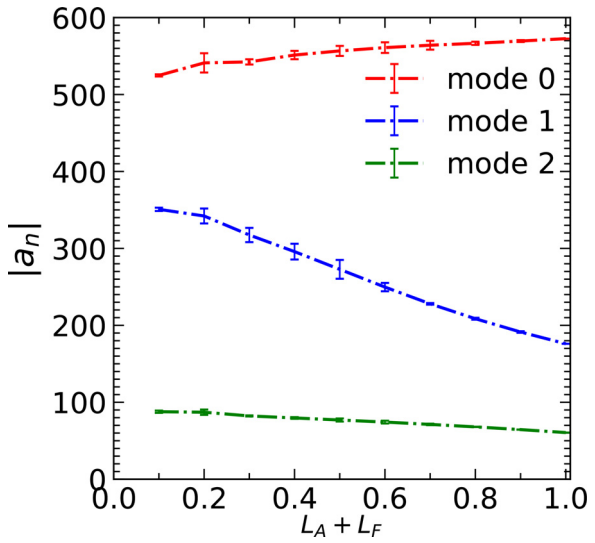


FIG. 13. Variation of DMD mode amplitudes $|a_n|$ of the modes 0, 1, and 2, with cylinder thickness, $L_A + L_F$. Errorbar indicates the range of variation of $|a_n|$ for different cross sections corresponding to the same $L_A + L_F$ values.

gradients in $L_{\overline{e'},a}$, leading to decrease in St with increase in vortex formation length. Similarly, the region of small $L_A + L_F$ is characterized with sharp gradients in θ_{sep}, leading to increase in St with downstream shifting of flow separation.

The vortex shedding plots indicate the vortex trajectories from either side approaching closest near the vortex formation point, β_{vf}. The downstream wake trajectory indicates a narrow 2S Kármán vortex shedding, with the β/α ratio approaching 0.281, as approximated by Milne-Thomson[75] using potential flow approximation. Later, the trajectories diverge and form a $C(2S)$ vortex shedding pattern, with $\beta/\alpha \in [0.54, 0.80]$. Finally, a quantitative measure of bluffness is proposed that is based on the fraction of pressure component in the total drag experienced by the cylinder. The bluffness definition is, qualitatively, in accordance with the previously used definition by Roshko.[66] The bluffness of cross sections approaching a vertical plate is 1, which is theoretically considered a perfectly bluff body. Furthermore, the proposed definition of bluffness is relatively easier for computations since only surface forces on the cylinder are needed.

While the present work reports results at $Re = 100$, it highlights fundamental insights into the role of frontbody and afterbody on the characteristics of fluid flow past a cylinder. These insights are useful to design several technical applications, namely, cross-section design of structural components undergoing flow loading, for particle capturing aquatic structures, improving shape optimization algorithms, and enhancing the efficiency of VIV based energy harvesting systems. The present study could be extended in future to study the joint effect of Re, L_F, and L_A on flow characteristics of a stationary cylinder and the effect of L_F and L_A on vortex-induced vibration of an elastically mounted cylinder.

ACKNOWLEDGMENTS

We gratefully acknowledge financial support by a grant (Grant No. MTR/2019/000696) from the Science and Engineering Research Board (SERB), Department of Science and Technology (DST), New Delhi, India.

AUTHOR DECLARATIONS

Conflict of Interest

The authors declare no known potential conflict of interest.

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

NOMENCLATURE

Symbol

a	Mode amplitude
C	Force coefficient per unit spanwise length, $2F^*/\rho_f^* u_0^{*2} D^*$
D^*	Cylinder diameter, m
$\overline{e'}_{max}$	Vortex strength, $0.5\,\overline{\mathbf{u}'\mathbf{u}'}_{max} = \overline{e'}^*_{max}/(0.5\rho_f^* u_0^{*2})$
f	Mode frequency
$L_{\overline{e'},a}$	Vortex formation length, $L^*_{\overline{e'},a}/D^*$
p	Local pressure, $p^*/\rho_f^* u_0^{*2}$
Re	Reynolds number, $\rho_f^* u_0^* D^*/\mu^*$
St	Strouhal frequency, $f_{v0}^* D^*/u_0^*$
t	Time, $t^* u_0^*/D^*$
$\mathbf{u}$	Local velocity vector, $(u/u_0^*,\ v/u_0^*)$
$\mathbf{u}'$	Local velocity fluctuation vector, $\bar{\mathbf{u}} - \mathbf{u}$
u_0^*	Free stream velocity (ms^{-1})
Φ	Mode shape

Superscripts

D	Drag force
L	Lift force
n	Mode number
rms	Root mean square value
$*$	Dimensional variable

Acronyms

FDM	Finite difference method
FEM	Finite element method
IB	Immersed boundary
GMRES	Generalized minimal residual

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